

Rate-Aware Decentralized Detection with Many Sensors

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Abstract—[Abstract placeholder.]

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I. INTRODUCTION

[Introduction placeholder.]

II. PROBLEM FORMULATION

Consider binary hypotheses H_1 and H_2 with priors $P(H_1) = p$ and $P(H_2) = 1 - p$. Sensor $i \in \{1, \dots, N\}$ observes $Y_i \in \mathcal{Y}$ and applies an encoder $\gamma_i : \mathcal{Y} \rightarrow \mathcal{U}$. Its message $U_i = \gamma_i(Y_i)$ is available to a fusion center (FC), which uses $\gamma_0 : \mathcal{U}^N \rightarrow \{1, 2\}$ to produce the final decision $U_0 = \gamma_0(U_{1:N})$.

Under uniform error cost,

$$c(H_j, U_0) = \mathbf{1}_{\{U_0 \neq j\}}, \quad J^N(\gamma_{0:N}) = \mathbb{E}^{\gamma_{0:N}}[c(H, U_0)]. \quad (1)$$

Thus, $J^N = P_e$ is the Bayes probability of error. We use the positive normalized error exponent

$$E_N(\gamma_{0:N}) = -\frac{1}{N} \log J^N(\gamma_{0:N}), \quad (2)$$

which is maximized when the error probability decays rapidly.

A. Assumptions

The analysis uses the following assumptions.

- 1) The observations $Y_1, \dots, Y_N$ are independent and identically distributed conditioned on the true hypothesis.
- 2) Under both hypotheses, the observation laws have densities $f_{Y|H}(y|H_j)$ with respect to a common reference measure.
- 3) All sensors have the same finite message alphabet $\mathcal{U}$.
- 4) The likelihood ratio

$$L(y) = \frac{f_{Y|H}(y|H_2)}{f_{Y|H}(y|H_1)} \quad (3)$$

is regular enough that threshold encoders are well defined; in particular, its derivative is continuous and nonsingular almost surely on the observation space.

- 5) The FC knows the sensor encoders, including any realized common randomization, as well as the received messages.

Under these conditions, the likelihood ratio is a sufficient statistic at each sensor. The results of [1] guarantee an optimal finite-population solution with threshold encoders and

a maximum a posteriori (MAP) decoder. They also show that identical independent sensor policies are optimal in the infinite-population problem and asymptotically optimal for large finite populations.

B. MAP fusion rule

For fixed deterministic encoders, define

$$\Delta_N(U_{1:N}) = \frac{1}{N} \sum_{i=1}^N \log \frac{P^{\gamma_i}(U_i|H_1)}{P^{\gamma_i}(U_i|H_2)}. \quad (4)$$

The MAP rule is

$$\gamma_0^*(U_{1:N}) = \begin{cases} 1, & \Delta_N(U_{1:N}) \geq t_N, \\ 2, & \Delta_N(U_{1:N}) < t_N, \end{cases} \quad t_N = \frac{1}{N} \log \frac{1-p}{p}. \quad (5)$$

III. ERROR-EXPONENT CHARACTERIZATION

For a sensor policy γ_i , define the policy-dependent Chernoff coefficient

$$M(\alpha, \gamma_i) = \sum_{u \in \mathcal{U}} P^{\gamma_i}(u|H_2)^{1-\alpha} P^{\gamma_i}(u|H_1)^\alpha, \quad 0 \leq \alpha \leq 1. \quad (6)$$

Applying Chernoff bounds to the two MAP error events gives

$$P_e \leq 2p^\alpha(1-p)^{1-\alpha} \prod_{i=1}^N M(\alpha, \gamma_i). \quad (7)$$

Suppose that the sensors use K distinct policies and that a fraction c_k uses policy γ_k , where $c_k \geq 0$ and $\sum_{k=1}^K c_k = 1$. From (7),

$$\liminf_{N \rightarrow \infty} E_N(\gamma_{0:N}) \geq -\min_{\alpha \in [0,1]} \sum_{k=1}^K c_k \log M(\alpha, \gamma_k). \quad (8)$$

For an identical policy γ , this becomes the Chernoff information between the two induced message distributions:

$$C_\gamma = -\min_{\alpha \in [0,1]} \log \left(\sum_{u \in \mathcal{U}} P^\gamma(u|H_2)^{1-\alpha} P^\gamma(u|H_1)^\alpha \right). \quad (9)$$

The structural result in [1] therefore reduces the large-network design to selecting one encoder that maximizes C_γ , subject here to a communication-rate constraint.

IV. RATE-AWARE GAUSSIAN POLICIES

Assume

$$Y_i|H_1 \sim \mathcal{N}(0, \sigma^2), \quad Y_i|H_2 \sim \mathcal{N}(\mu, \sigma^2), \quad (10)$$

with $\mu > 0$. Then

$$\log L(y) = \frac{\mu y}{\sigma^2} - \frac{\mu^2}{2\sigma^2}. \quad (11)$$

We compare an always-transmit binary policy with a confidence-based policy that can remain silent. Let $Q(x) = \int_x^\infty (2\pi)^{-1/2} e^{-z^2/2} dz$.

A. Always-transmit policy

For the symmetric threshold $\lambda_V = \mu/2$, let u_1 be sent below the threshold and u_2 otherwise. With $q = Q(\mu/(2\sigma))$, the induced probabilities are

$$\begin{aligned} P(u_1|H_1) &= P(u_2|H_2) = 1 - q, \\ P(u_2|H_1) &= P(u_1|H_2) = q. \end{aligned} \quad (12)$$

The Chernoff coefficient is symmetric about $\alpha = 1/2$, and hence

$$C_V = -\log \left(2\sqrt{q(1-q)} \right). \quad (13)$$

Every sensor transmits one bit, so the total rate is $R_V = N$ bits per observation interval.

B. Silent policy

Introduce a no-transmission symbol u_S and symmetric thresholds

$$\lambda_1 = \frac{\mu}{2} - \delta, \quad \lambda_2 = \frac{\mu}{2} + \delta, \quad \delta > 0. \quad (14)$$

The sensor sends u_1 for $Y_i < \lambda_1$, remains silent for $\lambda_1 \leq Y_i \leq \lambda_2$, and sends u_2 for $Y_i > \lambda_2$. Define

$$q_+ = Q\left(\frac{\mu/2 + \delta}{\sigma}\right), \quad q_- = Q\left(\frac{\mu/2 - \delta}{\sigma}\right). \quad (15)$$

The message probabilities are

$$\begin{aligned} (P(u_1|H_1), P(u_S|H_1), P(u_2|H_1)) &= (1 - q_-, q_- - q_+, q_+), \\ (P(u_1|H_2), P(u_S|H_2), P(u_2|H_2)) &= (q_+, q_- - q_+, 1 - q_-). \end{aligned} \quad (16)$$

Again, symmetry gives the minimizing Chernoff parameter $\alpha = 1/2$, so

$$C_S(\delta) = -\log \left(q_- - q_+ + 2\sqrt{(1 - q_-)q_+} \right). \quad (17)$$

The probability of transmission is $\rho(\delta) = 1 - q_- + q_+$ under either hypothesis. If an active sensor sends one bit and silence is detected at no bit cost, then

$$R_S(\delta) = N\rho(\delta) \quad (18)$$

is the expected total rate. Equations (17) and (18) expose the detection–communication tradeoff; δ can be selected by

$$\max_{\delta > 0} C_S(\delta) \quad \text{subject to} \quad R_S(\delta) \leq R_{\max}. \quad (19)$$

V. NUMERICAL EVALUATION

For the baseline experiment, use balanced priors and 0 dB SNR: $p = 1/2$ and $\mu = \sigma = 1$. Then $q = Q(0.5) \approx 0.3085$ and

$$C_V \approx 0.0793. \quad (20)$$

For likelihood-ratio thresholds $\tau_1 = 0.5$ and $\tau_2 = 2$, $\delta = \log 2$, $q_+ \approx 0.1164$, and $q_- \approx 0.5766$. Hence

$$C_S \approx 0.1007, \quad \rho \approx 0.5398. \quad (21)$$

This example predicts a larger asymptotic exponent while only about 54% of the sensors transmit. A complete evaluation should compare finite- N MAP error probabilities, the limits in (13) and (17), and the rate–exponent curve obtained by varying δ .

VI. CONCLUSION

The proposed structure combines the optimal-policy results for large sensor populations with a rate-aware encoder. The main remaining task is to optimize the silent region and verify the finite-population detection–communication tradeoff numerically.

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